

scan-no-removelast-seqcoll.sh

LVA-054 JDK21 SequencedCollection guard — a \$6.J anti-bluff source scanner.

Script: `scripts/scan-no-removelast-seqcoll.sh`

Purpose

Enforces that no production Kotlin uses the SequencedCollection-shaped `List/MutableList` methods that crash on Android runtimes below API 35.

The crash class (LVA-053 / LVA-054)

Kotlin `MutableList.removeLast()` / `removeFirst()` and `List.getFirst()` / `getLast()` compile, under JDK21 + AGP core-library-desugaring, to the JDK21 `java.util.SequencedCollection` methods (`java.util.List.removeLast`, etc.). Those methods do NOT exist on the Android platform's `java.util.ArrayList` below API 35, so a real user on Android 14 / 13 / 12 hits a runtime `NoSuchMethodError`. JVM unit tests pass (desktop `ArrayList` HAS the methods) — exactly the bluff this gate evicts: the test is green, the feature crashes for the user.

What it does

1. Greps tracked production Kotlin (*.kt, excluding test sources) for the forbidden `removeLast()` / `removeFirst()` / `getFirst()` / `getLast()` call shapes on `List/MutableList` receivers.
2. Exits 1 (printing the offending file:line) on any match; exit 0 when clean.

Use `.removeAt(lastIndex)` / `.first()` / `.last()` (the Kotlin-stdlib forms that do NOT desugar to `SequencedCollection`) instead.

Usage

`bash scripts/scan-no-removelast-seqcoll.sh`

Wiring

Invoked by `scripts/check-constitution.sh` (pre-push gate); hermetic test at `tests/check-constitution/test_no_removalast_seqcoll.sh`.

Constitutional bindings

§6.J (tests/gates must guarantee the product works for real users — this gate catches a runtime crash class invisible to JVM unit tests), §6.R-adjacent (mechanical source scanner).